

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: TBT3 Control Packet Routing
Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:

Remove Control Packet Routing Section from Chapter 13. There are no different requirements from a Router the supports TBT3 Compatibility when it comes to Control Packet routing.

Benefits as a result of the changes:

Make the spec less confusing as the spec today lists non-coherent requirements for Control Packet Routing when working in TBT3 Compatibility mode:
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| <ol style="list-style-type: none">1. It doesn't define when to use those rules2. It is expected that a Router would know its depth and Route String before enumerated3. It defines what to do for a Host Router and not for Device Router, while the whole 13.4 section is for Device Routers |
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An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
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None.

An analysis of the hardware implications:
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None.

An analysis of the software implications:
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None.

An analysis of the compliance testing implications:
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None.

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Actual Change

(a). 13.4.6 Control Packet Routing (Remove Section)

13.4.6 Control Packet Routing

13.4.6.1 Downstream-Bound Packets

~~A Router that receives a Control Packet with the *CM* bit set to 0b, shall route the packet according to the following rules:~~

- ~~• If the packet arrived on the Upstream Adapter, then:~~
 - ~~○ If the Router is a Host Router and the *TopologyID Valid* bit in Router Configuration Space is set to 0b, then the Router shall process the packet using the Uninitialized Router Flow in Section 13.4.6.2.~~
 - ~~○ Else the Router shall extract the Egress Adapter Number from the Route String that corresponds to the Router's depth in the Spanning Tree (as present in the *Depth* field in Router Configuration Space). For example, given a Route String with a TopologyID of 0,0,0,0,3,1,8 and a Router with a depth of 2, the Egress Adapter Number is 3.~~
 - ~~▪ If the extracted Adapter Number is 0, the Control Adapter of the Router shall consume the packet. The Router shall process the packet using the Enumerated Router Flow in Section 6.4.3.2.1.~~
 - ~~▪ If the extracted Adapter Number refers to a Protocol Adapter, the packet shall be dropped and the Router shall send the Connection Manager a Notification Packet with Event Code = *ERR_ADP* as defined in Table 6-12.~~
 - ~~▪ If the extracted Adapter Number refers to a disconnected or disabled Adapter, the Router shall drop the packet and shall send the Connection Manager a Notification Packet with Event Code = *ERR_CONN* as defined in Table 6-12.~~
 - ~~▪ If the extracted Adapter Number refers to a connected Adapter and the *Lock* bit in the Adapter Configuration Space is set to 1b, the Router shall drop the packet and shall send the Connection Manager a Notification Packet with Event Code = *ERR_LOCK* as defined in Table 6-12.~~
 - ~~▪ Else, the Router shall forward the packet over the Egress Adapter that matches the extracted Adapter Number.~~
- ~~• If the packet arrived on an Adapter that is not the Upstream Adapter, then:~~
 - ~~○ If the packet is an Inter-Domain Request or an Inter-Domain Response, then the Router shall modify the packet as follows, and then send the packet over the Upstream Adapter.~~
 - ~~▪ Replace the Route String in the packet with the Route String of the receiving Router within the receiving Domain, then add the Ingress Adapter Number of the Adapter connected to the Inter-Domain Link (see Figure 6-16).~~
 - ~~▪ Set the *CM* bit to 1b.~~

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

- ~~○ If the packet is a Read Request or a Write Request and the *TopologyID Valid* bit in Router Configuration Space is set to 1b, then the Router shall drop the packet and shall send the Adapter that originated the Request a Notification Packet with Event Code = *ERR_ENUM* as defined in Table 6-12.~~
- ~~○ If the Router is a Host Router and the packet is a Read Request or a Write Request and the *TopologyID Valid* bit in Router Configuration Space is set to 0b, then the Router shall drop the packet and shall send the Adapter that originated the Request a Notification Packet with Event Code = *ERR_NUA* as defined in Table 6-12.~~
- ~~○ Else, Router shall drop the packet and shall not send any packets in response.~~

~~13.4.6.2 Uninitialized Router Flow~~

~~If the packet is a Read Request or a Write Request that targets Router Configuration Space, the Router shall process the packet as described in Section 6.4.3.3.~~

~~Else, Router shall drop the packet and shall not send any packets in response.~~